

Scientific Computation

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Lecture 22 - Newton's Method for Multiple Variables

Lecture Outline

- Convergence of Newton's method
- Newton's method for nonlinear systems

- Multivariate optimization

Systems of Nonlinear Equations

Consider a system of n equations in n variables:

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(r) = 0$$

$$\text{Now: } F: \mathbb{R}^n \rightarrow \mathbb{R}^n, n \text{ equations}$$

$\uparrow$
 $n \text{ variables}$

$$\begin{cases} f_1(x_1, x_2, \dots, x_n) = 0 \\ \dots \\ f_n(x_1, \dots, x_n) = 0 \end{cases} \quad \begin{matrix} F(r) = 0 \\ r \in \mathbb{R}^n \end{matrix}$$

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Newton's Method for Systems

Use a multivariate linearization we have:

$$0 = F(x^*) \approx \bar{F}(x^{(k)}) + J(x^{(k)})(x^* - x^{(k)})$$

where

ref. .

$$J(x) \triangleq D_x F(x)$$

$$J(x^{(k)})_{:j} \triangleq \left. \frac{\partial f_i}{\partial x_j} \right|_{x=x^{(k)}} \quad \downarrow \text{Jacobian operator}$$

is the Jacobian matrix of F . Define $S \triangleq x^{(k+1)} - x^{(k)}$ as the step from iteration k to $k+1$, we get the following linear system:

$$J(x^{(k)})S = -F(x^{(k)})$$

$$x^{(k+1)} = x^{(k)} + S$$

S is step

and we back to solve linear systems.

$$J(x) = \begin{bmatrix} \nabla f_1(x)^T \\ \vdots \\ \nabla f_n(x)^T \end{bmatrix}$$

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Linear Systems Return

We have reduced the problem of solving nonlinear systems of equations to solving a sequence of linear systems of equations:

$$J(x^{(k)}) S^{(k)} = -F(x^{(k)})$$

$$x^{(k+1)} = x^{(k)} + S^{(k)}$$

∇ gradient is used for scalar multi variable function

J Jacobian is used for vector multivariable function

H Hessian is used for scalar multi function for "second derivative"
-var

$$H = J(\nabla f)$$

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A Simple Example

A Simple Example

Example 1: Consider the nonlinear system

$$0 = F(x) = \begin{bmatrix} x_1^2 + x_2^2 - 25 \\ x_1^2 - x_2 - 1 \end{bmatrix}.$$

The Jacobian is:

$$J(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 & 2x_2 \\ 2x_1 & -1 \end{bmatrix}$$

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One Iteration of Newton's Method

Example 1 cont.: Let's use this Jacobian to perform the first iteration of Newton's method for $x_0 = [5 \ 1]^T$:

$$F(x_0) = \begin{bmatrix} 5^2 + 1^2 - 25 \\ 5^2 - 1 - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 23 \end{bmatrix}$$

$$J(x_0) = \begin{bmatrix} 2 \cdot x_1^{(0)} & 2 \cdot x_2^{(0)} \\ 2 \cdot x_1^{(0)} & -1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 5 & 2 \cdot 1 \\ 2 \cdot 5 & -1 \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ 10 & -1 \end{bmatrix}$$

Solve $\begin{bmatrix} 10 & 2 \\ 10 & -1 \end{bmatrix} S = \begin{bmatrix} -1 \\ -23 \end{bmatrix}$

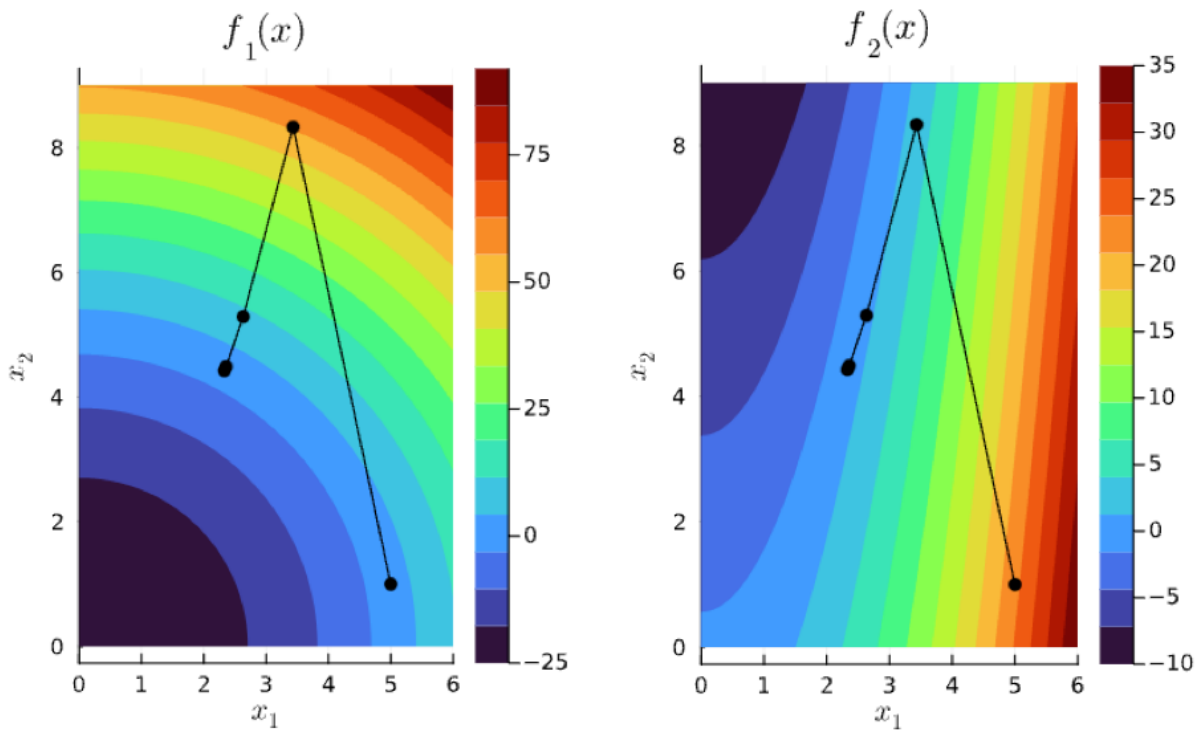
non-singular
 $\Rightarrow$ unique solution

$\rightarrow X^{(k+1)} - X^{(k)}$
(step we take)

$$S = \begin{bmatrix} -1.6 \\ -7 \end{bmatrix}, \quad X^{(1)} = X^{(0)} + S^{(0)} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} + \begin{bmatrix} -1.6 \\ -7 \end{bmatrix} = \begin{bmatrix} 3.4 \\ -6 \end{bmatrix}$$

Visualizing 2D Systems

Example 1 cont.: Let's examine contour maps for the components of $F(x)$:



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Accompanying Code

Which can be accomplished with a few lines of code:

```
using Plots
x0 = [5; 1]
f1(x1, x2) = x1^2 + x2^2 - 25.0
f2(x1, x2) = x1^2 - x2 - 1.0
F(x1, x2) = [f1(x1, x2); f2(x1, x2)]
J(x1, x2) = [2*x1 2*x2;
              2*x1 -1.0]

function newton(x, F, J)
    s = -J(x[1], x[2])\F(x[1], x[2])
    return x + s
end
```

```
N_plot = 500
X1 = range(0.0, stop=6.0, length=N_plot)
X2 = range(0.0, stop=9.0, length=N_plot)
z1 = @. f1(X1, X2)
contourf(X1, X2, z1, fill=true, color=:turbo, lw=0.0,
          aspect_ratio=:equal)
plot!(X1_newton, X2_newton, color=:black, primary=false)
scatter!(X1_newton, X2_newton, color=:black, primary=false)
axis!(0.0, 6.0)
axis!("\$x_1\$")
axis!("\$x_2\$")
```

```

end

x1 = newton(x0, F, J)
x2 = newton(x1, F, J)
x3 = newton(x2, F, J)
x4 = newton(x3, F, J)

X1_newton = [x0[1]; x1[1]; x2[1]; x3[1]; x4[1]]
X2_newton = [x0[2]; x1[2]; x2[2]; x3[2]; x4[2]]

title!("\$f_1(x)\$")

z2 = @. f2(X1', X2)
contourf(X1, X2, z2, fill=true, color=:turbo, lw=0.0,
         aspect_ratio=:equal)
plot!(X1_newton, X2_newton, color=:black, primary=false)
scatter!(X1_newton, X2_newton, color=:black, primary=false)
xaxis!([0.0, 6.0])
xaxis!("\$x_1\$")
yaxis!("\$x_2\$")
title!("\$f_2(x)\$")

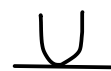
```

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Lecture Outline

- Convergence of Newton's method
- Newton's method for nonlinear systems
- **Multivariate optimization**

Newton's Method for Multivariate Optimization



Let's apply Newton's method to multivariate optimization problems. In the scalar case, we had $f'(x) = 0$ defining stationary points. For a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the necessary optimality condition is

$$\nabla f(x) = 0$$

$$\nabla f(x) = 0.$$

$$\nabla f: \mathbb{R} \rightarrow \mathbb{R}$$

To find roots of ∇f , Newton's method needs its Jacobian

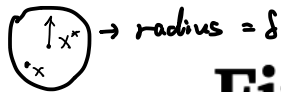
which is actually the Hessian $\nabla^2 f$

$$\nabla^2 f = H(f) = J(\nabla f)$$

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} \Rightarrow D_x \nabla f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \frac{\partial^2 f}{\partial x_2^2} & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \dots & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \triangleq \nabla^2 f(x)$$

$$\Rightarrow \nabla^2 f(x^{(k)}) (x^{(k+1)} - x^{(k)}) = -\nabla f(x^{(k)})$$

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First Order Optimality

We have mentioned that $\nabla f(x) = 0$ is a necessary optimality condition — let's quickly prove this!

Let x^* be a local minimum: i.e. there exists $\delta \geq 0$, s.t. if $\|x - x^*\|_2 \leq \delta$, then $f(x^*) \leq f(x)$. Write the Taylor series around x^* : $f(x) = f(x^*) + \nabla f(x^*)^T (x - x^*) + O(\|x - x^*\|_2^2)$.

Select x s.t. $x - x^* = -\alpha \nabla f(x^*)$, where $\alpha > 0$. If we choose α small enough, $\nabla f(x^*)^T (x - x^*) = -\alpha \|\nabla f(x^*)\|_2^2 < 0$, and $-\alpha \|\nabla f(x^*)\|_2^2$ has magnitude more than the quadratic terms because $\|x - x^*\|_2^2 = \alpha^2 \|\nabla f(x^*)\|_2^2$,
 $\Rightarrow f(x) < f(x^*)$ unless $\nabla f(x^*) = 0$.

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Sufficient Conditions for Local

Sufficient Conditions for Local

Minima

$$\nabla^2 f_S = -\nabla f$$

The necessary first order optimality condition is $\nabla f(x) = 0$, but this is true for all *critical points* (minima, maxima, and saddle points). To check whether we've found a local minimum:

Theorem 1: A critical point x^* of $f(x)$ is a **local minimum** if $\nabla^2 f(x^*) \succ 0$ (that is, its **Hessian at x^* is positive definite**).

$A \in \mathbb{R}^{n \times n}$ is positive definite if $x^T A x > 0$ for all $x \neq 0$

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Proof of Sufficient Conditions for Local Minima

This result is easily proved by examining the second-order Taylor series approximation around x^* :

$$f(x) = f(x^*) + \cancel{\nabla f(x^*)}^T (x - x^*) + \frac{1}{2} (x - x^*)^T \nabla^2 f(x^*) (x - x^*) + O(\|x - x^*\|^3)$$

3rd order and higher polynomial terms

First order critical point is necessary for minimum: $\nabla f(x^*) = 0$

$f(x) > f(x^*)$ in a neighbourhood of x^* if $\nabla^2 f(x^*) \succ 0$ (i.e. $\nabla^2 f$ is PD)

A Simple Optimization Example

Example 2: Consider the nonlinear unconstrained optimization problem

$$\min_{x \in \mathbb{R}^2} f(x)$$

where $f(x) = \frac{1}{4}(x_1 - 1)^4 + (x_1 x_2 - \frac{1}{2})^2 + 1.0$.

We want roots of

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} (x_1 - 1)^3 + 2x_2(x_1 x_2 - \frac{1}{2}) \\ 2x_1(x_1 x_2 - \frac{1}{2}) \end{bmatrix}$$

Critical points = 0

Newton's Method for Example 2

Example 2 cont.: The Hessian of this function is:

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

$$= \begin{bmatrix} 3(x_1 - 1)^2 + 2x_2 & 2x_1 \\ 2x_1 & 2x_1^2 - 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5(x_1-1) + 2x_2 & x_1x_2 - \frac{1}{2} \\ 4x_1x_2 - 1 & 2x_1^2 \end{bmatrix}$$

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Solving Example 2 by Hand

Example 2 cont.: We can find the solution for this toy problem by inspection and check the sufficient optimality conditions:

$$f(x) = \underbrace{\frac{1}{4}(x_1-1)^4}_{\geq 0} + \underbrace{(x_1x_2 - \frac{1}{2})^2}_{\geq 0} + 1 \geq 0$$

$\Rightarrow f(x)$ has a lowest possible value of

$f(x^*) = 1$, check $x_1 - 1 = 0 \Rightarrow x_1^* = 1$

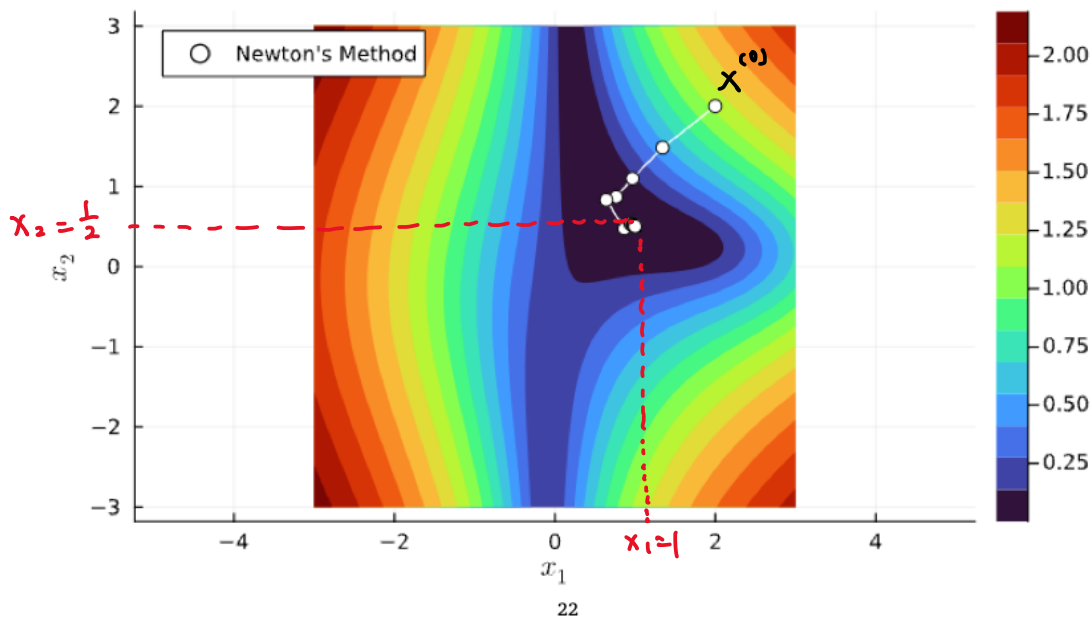
$$x_1^* x_2 = \frac{1}{2} \Rightarrow x_2^* = \frac{1}{2}$$

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Example 2 Visualized

Here is a plot of Newton's method initialized with $x^{(0)} = [2 \ 2]^T$ (note that the cost function is passed through $\log_{10}$ for effect):

$$\log_{10}(f(x))$$



Eigenvalues of $\nabla^2 f$

Recall that a positive definite (PD) Hessian matrix $\nabla^2 f$ is sufficient for a critical point of f to be an unconstrained minimizer.

Simple test for positive definiteness:

Get the smallest (algebraically) eigenvalue of $D^2 f(x)$, check that it's positive, if so, $D^2 f(x) \succ 0$

we know this because PD $\Rightarrow$ all eigen values are positive

多维牛顿法还可以做多元优化问题：

一、目标问题：多元函数的最优化

我们现在不是在解方程 $f(x) = 0$ ，
而是在找极值点（最小值或最大值）：

$$\min f(x) \quad \text{或} \quad \max f(x)$$

对于多元函数 $f: \mathbb{R}^n \rightarrow \mathbb{R}$ ，
极值点的必要条件是：

$$\nabla f(x) = 0$$

也就是说：

只要梯度为零（即所有偏导都为零），该点就是一个驻点（stationary point）。

二、连接到 Newton's Method

Newton's 方法原本是用来找方程的根：

$$f(x) = 0$$

现在我们让 Newton's 方法作用在：

$$\nabla f(x) = 0$$

（也就是“让梯度函数当作目标函数”）

于是，我们就把优化问题转化为「找梯度的根」。

三、关键：Jacobian of the gradient = Hessian

对于多元函数 $f(x_1, \dots, x_n)$ ，它的梯度是：

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Newton's 方法要求我们计算梯度的 Jacobian：

$$D_x(\nabla f(x)) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

这个矩阵，就是熟悉的 **Hessian (海森矩阵)**：

$$H(x) = \nabla^2 f(x)$$

✿ 四、Newton's 更新公式 (在优化中)

Newton's 方法用于优化时，就变成了：

$$H(x^{(k)}) (x^{(k+1)} - x^{(k)}) = -\nabla f(x^{(k)})$$

或者写作：

$$x^{(k+1)} = x^{(k)} - [H(x^{(k)})]^{-1} \nabla f(x^{(k)})$$

🕒 五、几何意义

- $\nabla f(x)$: 告诉你「梯度方向」—— 函数上升最快的方向。
- $H(x)$: 告诉你「曲率」—— 函数在不同方向上弯曲的程度。

Newton's 方法用二阶近似 (梯度 + 曲率) 去**预测最优点在哪**，
比单纯用梯度下降快得多。

Heissan 告诉你函数在不同的地方函数的弯曲程度，而gradient告诉你函数上升最快的方向。

✿ 一、问题背景：什么是一阶最优性条件？

我们在做优化 (minimize $f(x)$) 时，

如果 x^* 是一个 **局部最小点 (local minimum)**，

那意味着在它周围一个小范围内：

$$f(x^*) \leq f(x) \quad \text{for all } x \text{ near } x^*.$$

这时我们要证明：

👉 “这会强制使得梯度必须为 0”。

🧠 二、证明思路 (利用泰勒展开)

在点 x^* 附近, 对 $f(x)$ 做二阶泰勒展开:

$$f(x) = f(x^*) + \nabla f(x^*)^T(x - x^*) + O(\|x - x^*\|^2)$$

意思是:

函数在 x^* 附近的变化, 主要由梯度项 (线性部分) 决定。

■ 三、反证法逻辑

假设 x^* 是局部最小点, 但梯度不是零 ($\nabla f(x^*) \neq 0$) 。

那我们可以沿着 **负梯度方向** (下降最快的方向) 走一点:

$$x = x^* - \alpha \nabla f(x^*)$$

其中 $\alpha > 0$ 很小。

⚙ 四、代入展开式

代入泰勒展开:

$$f(x) = f(x^*) + \nabla f(x^*)^T(x - x^*) + O(\|x - x^*\|^2)$$

由于:

$$x - x^* = -\alpha \nabla f(x^*)$$

所以:

$$\nabla f(x^*)^T(x - x^*) = -\alpha \|\nabla f(x^*)\|^2 < 0$$

而 $O(\|x - x^*\|^2)$ 是非常小的正数项 (因为 α 很小) ,
所以线性项主导。

于是:

$$f(x) \approx f(x^*) - \alpha \|\nabla f(x^*)\|^2 < f(x^*)$$

这就说明:

你可以找到一个点 x , 使 $f(x)$ 比 $f(x^*)$ 更小。

这违反了“局部最小点”定义。



✅ 五、结论

所以:

如果 x^* 是局部最小点, 则 $\nabla f(x^*) = 0$.

也就是说, **梯度为零是局部最优的必要条件**。

✿ 一、背景：一阶条件只能找“临界点”

上一页我们已经证明了：

如果 x^* 是局部最小点 $\rightarrow$

$$\nabla f(x^*) = 0$$

但问题是：

$\nabla f(x^*) = 0$ 这个条件对所有 **极值点和鞍点** 都成立, 比如：

- 局部最小值 ✓
- 局部最大值 ✓
- 鞍点 (saddle point) ✓

所以这个条件只是 **必要** (necessary) ,
但还**不充分** (not sufficient) 。

🧠 二、我们需要“二阶信息”来区分这些情况

为了判断它是哪种点, 我们要看函数在 x^* 附近的“弯曲方向”——这就靠 **Hessian 矩阵**。

Hessian:

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

⚙️ 三、二阶充分条件 (sufficient condition)

定理 1:

若 x^* 是临界点 (即 $\nabla f(x^*) = 0$) ,
且 Hessian $\nabla^2 f(x^*)$ 是**正定的** (positive definite) ,
则 x^* 是一个局部最小点。

📏 四、什么叫“正定矩阵”

定义如下：

$A \in \mathbb{R}^{n \times n}$ 是正定的 (positive definite) 当且仅当 $x^T A x > 0, \forall x \neq 0$

直觉解释：

符号

含义

几何意义

$x^T A x > 0$	“方向上的曲率为正”	所有方向上函数都向上弯（像碗底）
$x^T A x < 0$	“方向上的曲率为负”	所有方向上函数都向下弯（像山顶）
混合正负	部分方向上弯上、部分弯下	鞍点 (saddle point)

📊 五、几何解释（直觉）

- **正定 Hessian** → **最小值**

二阶导都正，函数像一个“碗口向上”的曲面。

所以 x^* 是局部 minimum。

- **负定 Hessian** → **最大值**

二阶导都负，曲面“碗口向下”。

- **不定 Hessian** → **鞍点**

有的方向弯上，有的方向弯下。

✅ 六、总结表格

条件	结果	形状比喻
$\nabla f(x^*) = 0, \nabla^2 f(x^*) \succ 0$	局部最小值	碗底 🍲
$\nabla f(x^*) = 0, \nabla^2 f(x^*) \prec 0$	局部最大值	山顶 🏔️
$\nabla f(x^*) = 0, \nabla^2 f(x^*)$ 不定	鞍点	马鞍 🐎

雅可比矩阵是干啥的：

相比于gradient，雅可比矩阵用在多输入多输出的情况（多元向量函数）；它描述一个局部的线性变换，即输入向量为微小的变化delta的时候，输出会如何变化。Jacobian是雅可比矩阵的determinant，用来描述坐标变换的面积变化。